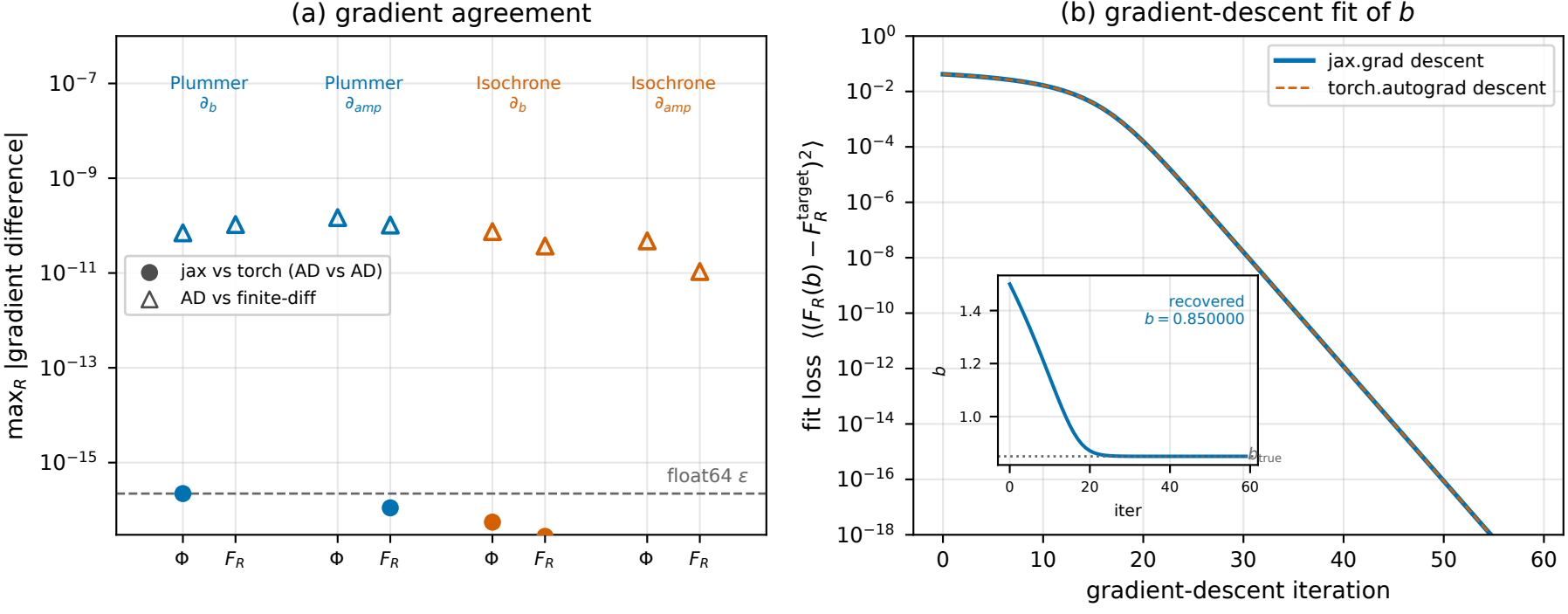
(a) gradient agreement
Plummer
∂_b
Plummer
∂_{amp}
Isochrone
∂_b
Isochrone
∂_{amp}
jax vs torch (AD vs AD)
AD vs finite-diff
float64 ε
$\max_R$ |gradient difference|
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
Φ
F_R
(b) gradient-descent fit of b
jax.grad descent
torch.autograd descent
recovered
$b = 0.850000$
b_{true}
iter
fit loss $\langle (F_R(b) - F_R^{\text{target}})^2 \rangle$
gradient-descent iteration